



Place, Perception, and Narrative in Film Data
Spatial and Textual Analysis of Violence and Poverty Themes
MSc Data Science | Spatial + Text Analysis. Middlesex University 2026

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? Research Question:

Are violence-related and poverty-related themes more prevalent in films produced in the Southern Hemisphere compared to the Northern Hemisphere (1990–2024)?

Method

Transformer embeddings
Semantic similarity
Spatial aggregation

Data

Source: IMDb film summaries
Features: -Text (summaries)
- Country of origin
-Hemisphere classification

index	Label Name	What it Represents
0	violence_flag	Binary indicator (1 or 0) for the presence of violence-related themes in a film's synopsis.
1	poverty_flag	Binary indicator (1 or 0) for the presence of poverty-related themes in a film's synopsis.
How it was Created		
Initially derived by checking for keywords. Later refined using Sentence Transformer cosine similarity scores (violence_score) and a threshold of 0.30.		
Initially derived by checking for keywords. Later refined using Sentence Transformer cosine similarity scores (poverty_score) and a threshold of 0.30.		

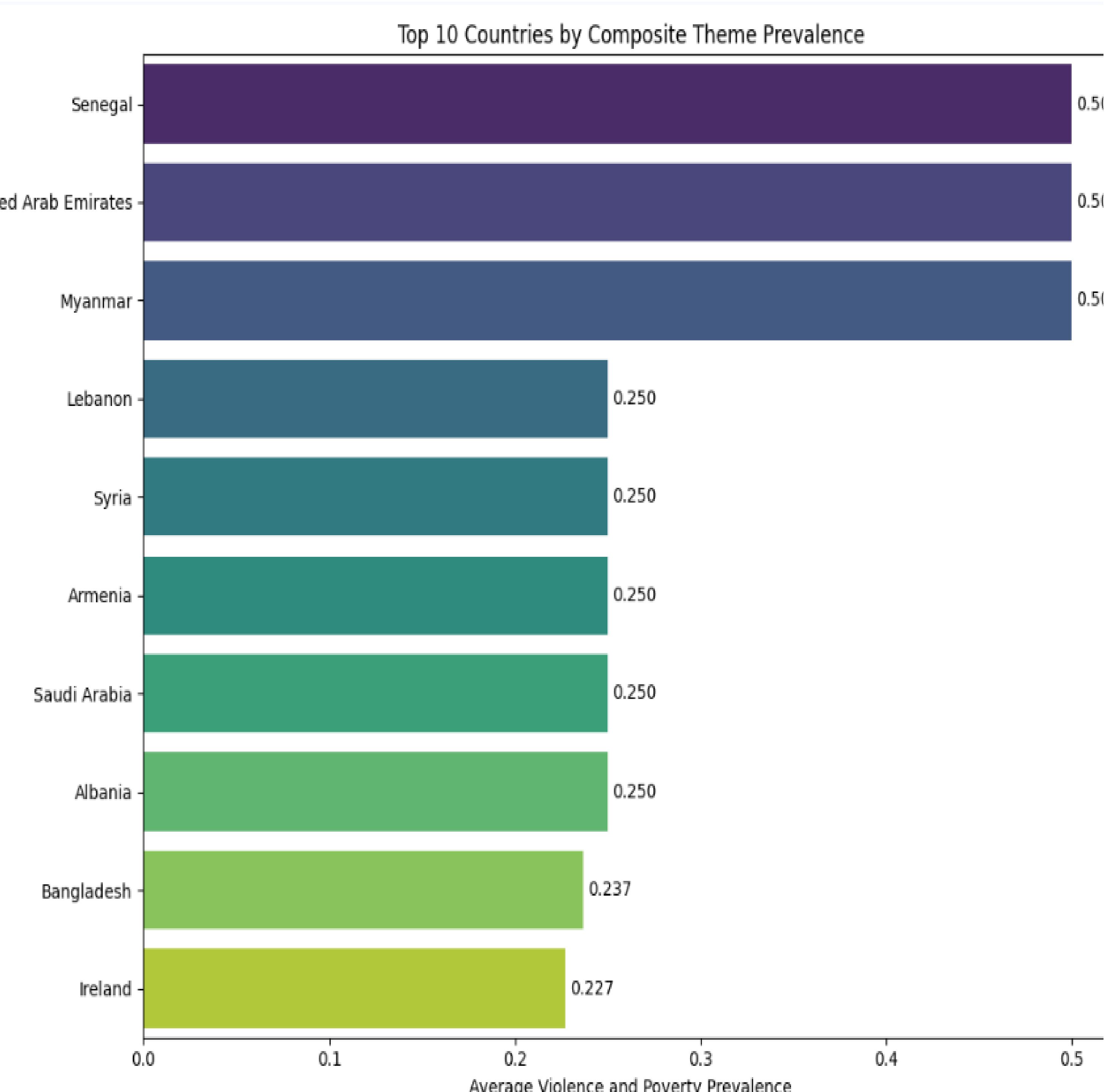
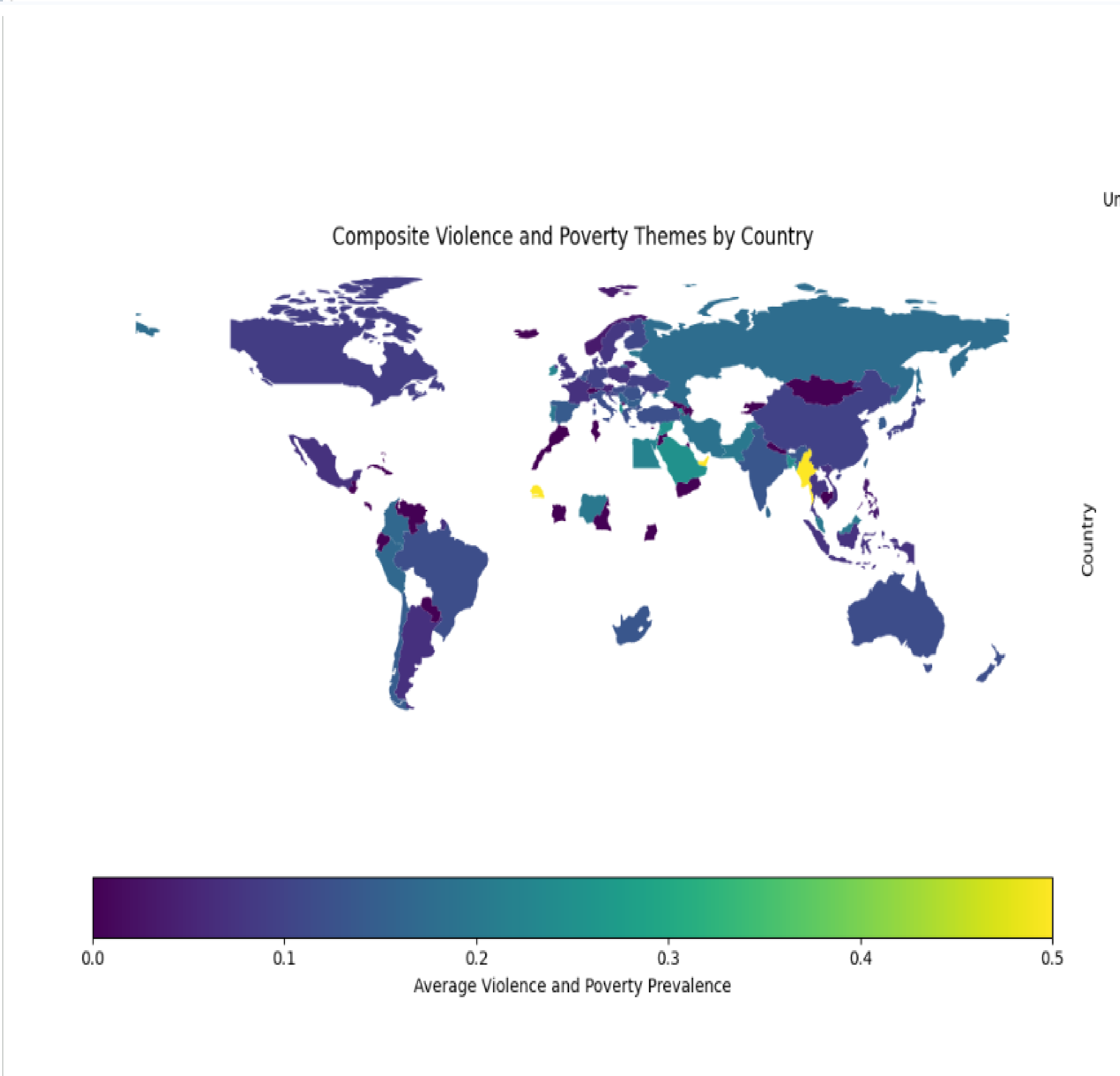
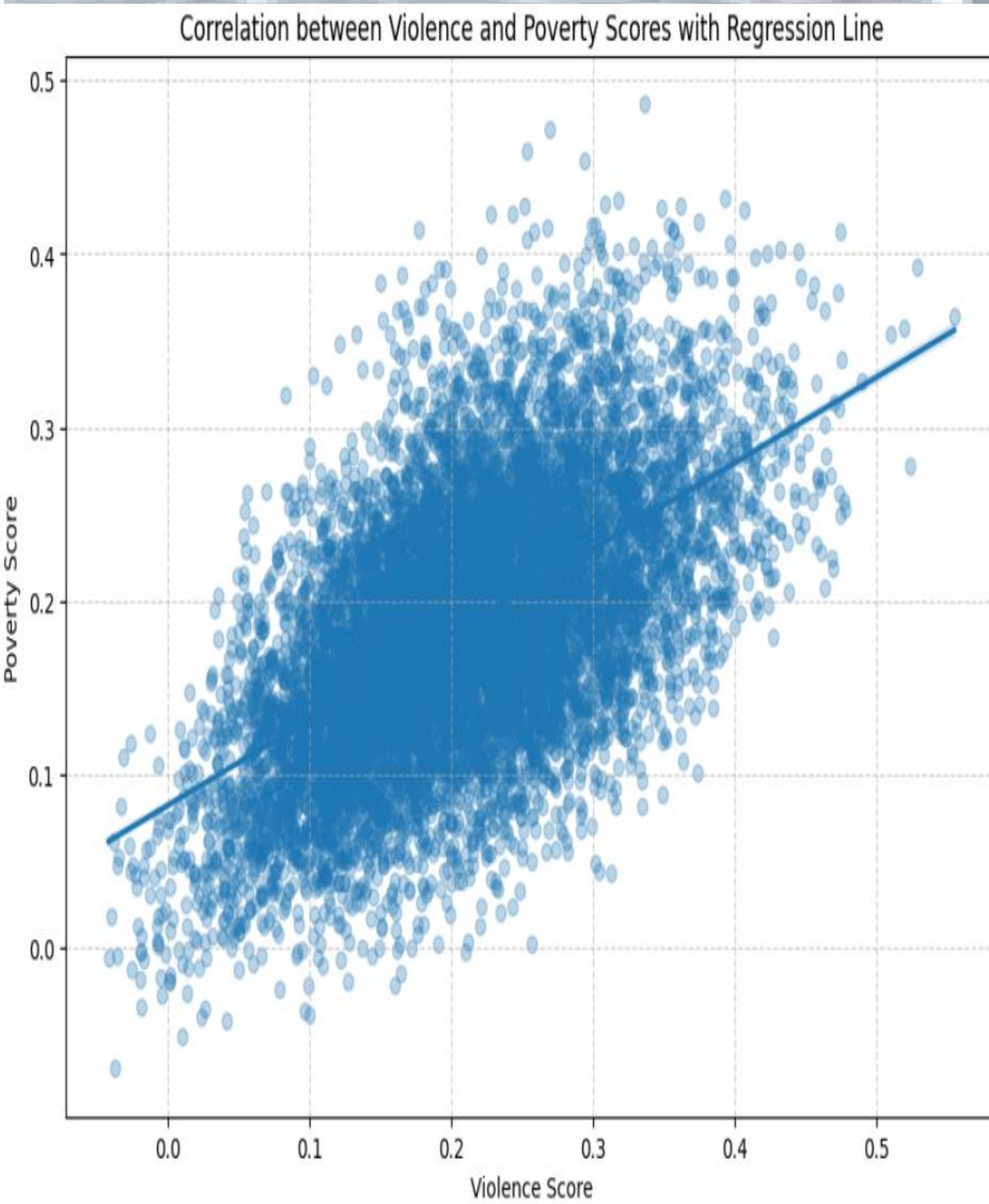
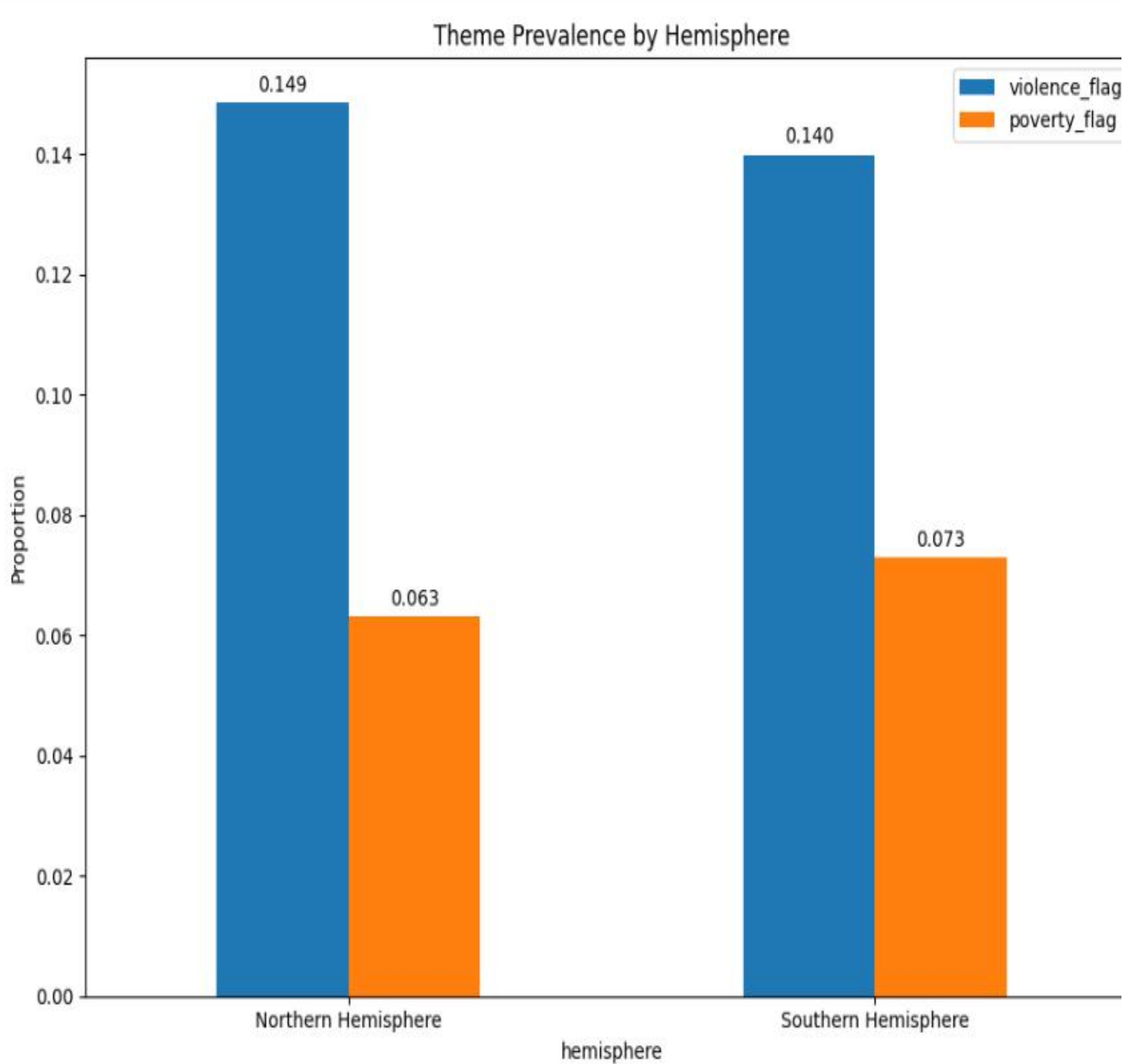
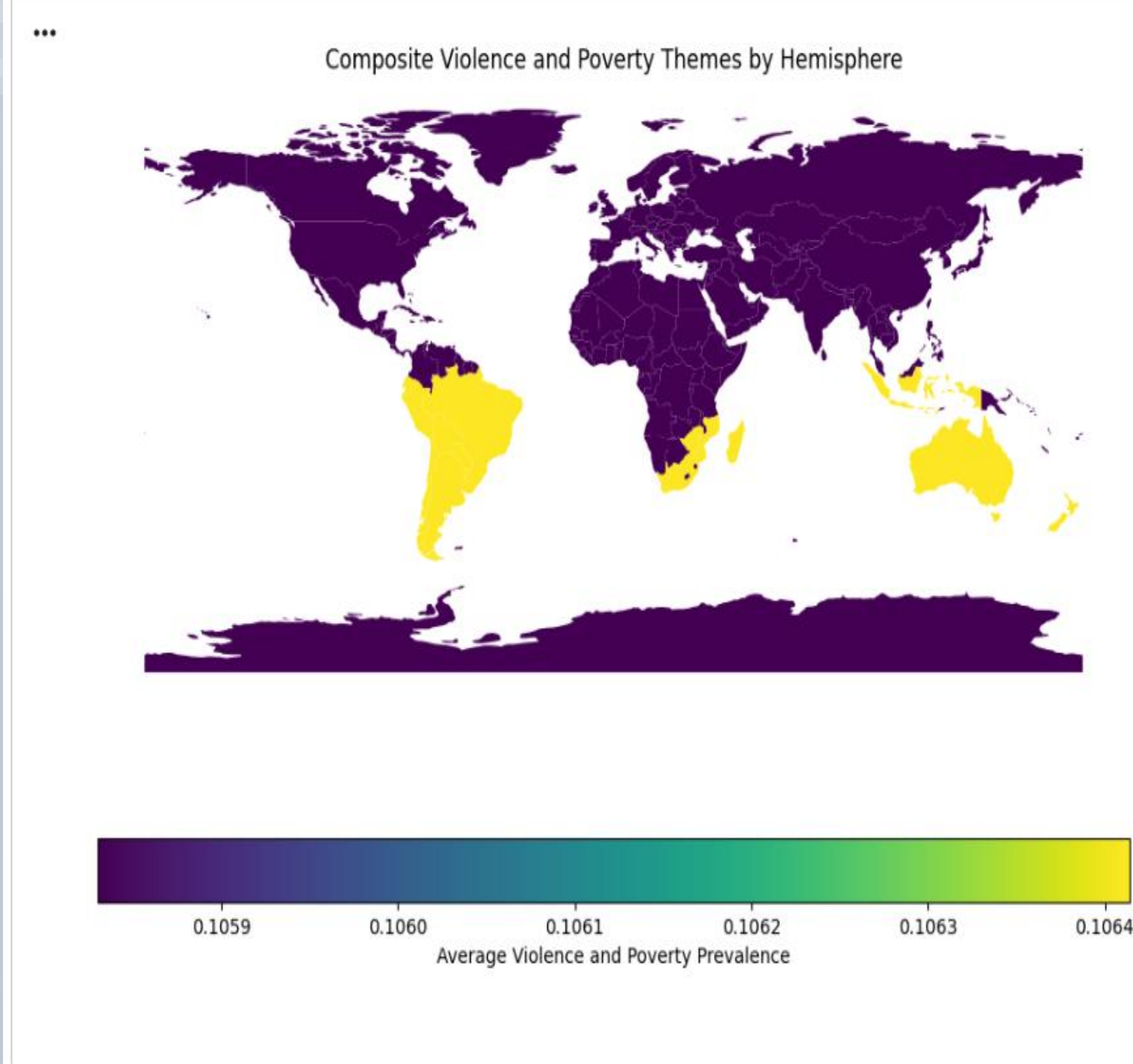
Limitation

Data imbalance
- No ground truth
- Threshold sensitivity

No clear geographic clustering.

Small differences between hemispheres

Themes co-occur (positive relationship)



Themes show variation across countries. No clear geographic clustering is observed.

Narrative themes are not geographically determined

NLP improves theme detection, but geographic patterns remain weak.

- Themes are continuous, interconnected, and not spatially structured.

Violence-related and poverty-related themes tend to co-occur within film narratives.

Films with higher violence scores are more likely to also exhibit higher poverty-related signals.

References

-IMDb (2024) IMDb Datasets. Available at: <https://datasets.imdbws.com/> (Accessed: 31 March 2026).
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-Manning, C.D., Raghavan, P. and Schütze, H. (2008). Introduction to Information Retrieval.
-Tobler, W. (1970) 'A computer movie simulating urban growth in the Detroit region', Economic Geography.